OpenVPN –

What is OpenVPN?

OpenVPN is an open-source VPN solution used to create a **secure encrypted tunnel** between devices over the Internet.

It allows users to securely access:

- Company networks
- Remote servers
- Private resources

OpenVPN uses:

- SSL/TLS encryption
- Certificates for authentication
- UDP or TCP transport protocols

Why Use OpenVPN?

Without VPN:

```
Laptop ---- Internet ---- Server
```

Data may be intercepted.

With OpenVPN:

```
Laptop ==== Encrypted Tunnel ==== Server
```

All traffic is encrypted.

How OpenVPN Works

Step 1

Client connects to OpenVPN Server.

Step 2

Server and Client authenticate each other.

Step 3

Encryption keys are exchanged.

Step 4

Secure tunnel is established.

Step 5

All traffic passes through the encrypted tunnel.

OpenVPN Components

1. OpenVPN Server

Provides VPN service.

Example:

```
Ubuntu Server
Windows Server
CentOS Server
```

---

2. OpenVPN Client

Connects to the VPN server.

Examples:

- Windows Client
- Linux Client
- Android Client
- iPhone Client

---

3. Certificate Authority (CA)

Issues certificates for authentication.

```
CA
    Server Certificate
    Client Certificate
```

OpenVPN Modes

1. Routed Mode (TUN)

Works at Layer 3.

Uses IP packets.

```
Client ---- VPN Tunnel ---- Server
```

Virtual Interface:

```
tun0
```

Most commonly used.

---

2. Bridged Mode (TAP)

Works at Layer 2.

Uses Ethernet frames.

```
Client ---- TAP Tunnel ---- Server
```

Virtual Interface:

```
tap0
```

Used when broadcast traffic is needed.

OpenVPN Ports

Default:

```
UDP 1194
```

Can also use:

```
TCP 1194
```

OpenVPN Authentication Methods

Username & Password

```
User: admin
Password: ********
```

Certificate-Based Authentication

Most secure.

```
Client Certificate
Server Certificate
CA Certificate
```

Multi-Factor Authentication (MFA)

```
Password + OTP
```

OpenVPN Topologies

1. Remote Access VPN

One user connects to office network.

```
Laptop
   ||
OpenVPN
   ||
Office Network
```

---

2. Site-to-Site VPN

Connects two offices.

```
Office A
   ||
OpenVPN Tunnel
   ||
Office B
```

OpenVPN Encryption

Common algorithms:

- AES-128
- AES-256
- SHA256
- SHA512

Example:

```
Cipher: AES-256-GCM
Auth: SHA256
```

OpenVPN Configuration Files

Server

```
/etc/openvpn/server.conf
```

Client

```
client.ovpn
```

Contains:

- Server IP
- Port
- Certificates
- Encryption settings

Basic Packet Flow

```
Client
   |
Encrypt
   |
Internet
   |
Decrypt
   |
Server
```

OpenVPN Advantages

- Free and Open Source
- Strong Security
- Cross Platform
- Supports NAT
- Easy Remote Access
- Works Behind Firewalls

OpenVPN vs IPSec

Feature	**OpenVPN**	**IPSec**
Security	High	High
Setup	Easier	More Complex
Firewall Traversal	Better	Sometimes Difficult
Port	UDP/TCP 1194	UDP 500/4500
Authentication	SSL/TLS	IKE
Open Source	Yes	Standard Protocol

OpenVPN is an open-source VPN solution that uses **SSL/TLS encryption** to create a secure tunnel between a client and a server. It supports both **UDP and TCP**, uses certificates for authentication, and is commonly used for **remote access VPN** and **site-to-site VPN** connections. By default, OpenVPN uses **UDP port 1194** and provides secure encrypted communication over public networks.